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SCSE 2123 SOFTWARE REQUIREMENTS ENGINEERING

TOPIC 8: Practices for Requirements Validation and Negotiation

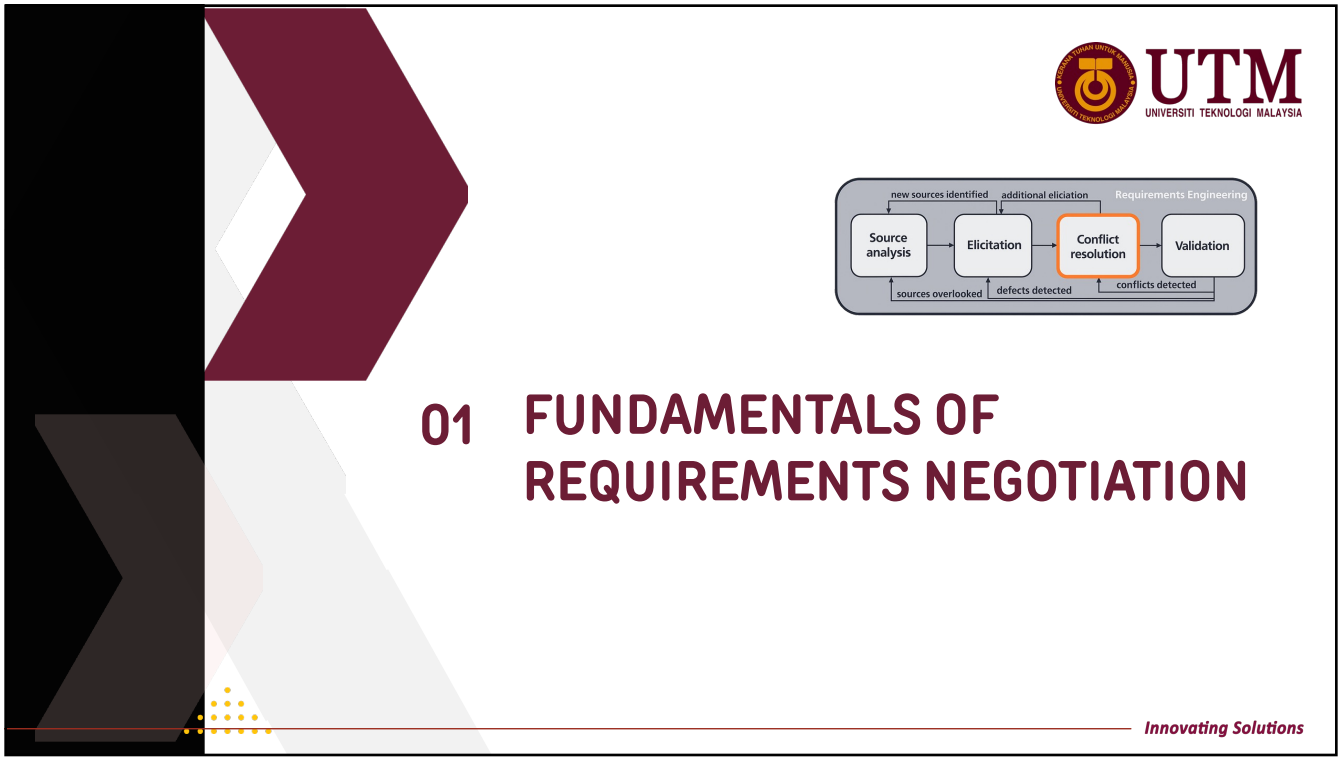
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Contents

- Fundamentals of Requirements Negotiation
- Fundamentals of Requirements Validation

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Requirements Conflict

REQUIREMENTS NEGOTIATION

Definition

A conflict in requirements engineering exists, if the needs and wishes of different stakeholders (or group of stakeholders) regarding the system contradict each other.

- All stakeholders must understand and agree on all requirements that are relevant to them for the final set of requirements.
- If some stakeholders do not agree, this situation of conflict should be resolved.
- Conflict resolution techniques need to be selected based on the type of conflict and contextual information.
- Requires in-depth understanding of the nature of the requirements conflict and of the attitudes of the stakeholders involved.

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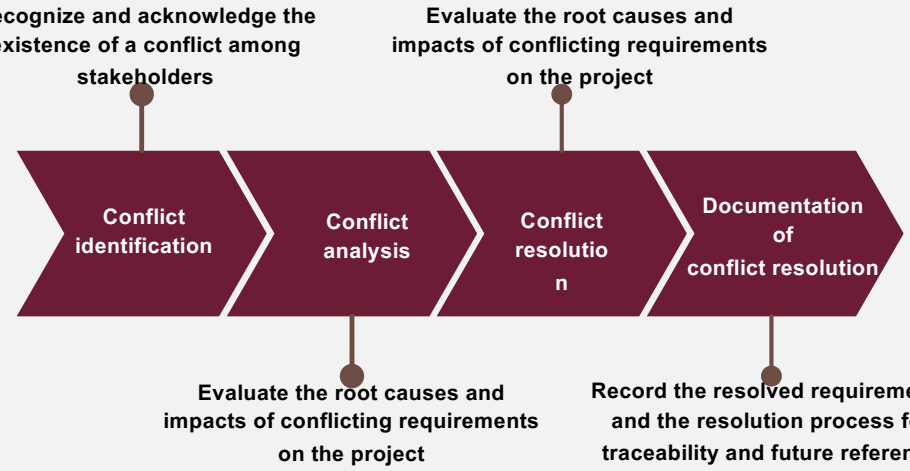
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Conflict Management Activities

REQUIREMENTS NEGOTIATION



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Conflict Analysis: Type of Conflicts

REQUIREMENTS NEGOTIATION

1. Subject Matter Conflict

- Stakeholders have **different factual needs**, mostly caused by the intended use of he system in different environments.

2. Data Conflict

- Stakeholders interpret
  - given **information differently**
  - have **inconsistent data** from different sources
  - different views on what is relevant
  - different assessment procedure

3. Interest Conflict

- Stakeholders have **different goals or interests** (personal, related to group or role)
- Can be objective or subjective in nature

4. Value Conflict

- Stakeholders have
  - **divergent values and principles**
  - **different criteria for evaluation ideas or behaviour**

5. Relationship Conflict

- There are **emotional problems** in personal relationship between stakeholders.
- negative interpersonal behaviour between stakeholders

6. Structural Conflict

- Conflict involves
  - **inequality of power or authority**,
  - competition over limited resources, or structural dependencies between parties.

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Conflict Analysis: Type of Conflicts – Examples

REQUIREMENTS NEGOTIATION

1. Subject Matter Conflict

- The **Logistics Manager** requires the system to calculate shipping costs based on the physical dimensions of the box to optimize truck space. However, the **Marketing Manager** requires a flat-rate shipping fee to simplify the checkout process for customers

2. Data Conflict

- The **Finance Team** interprets "Monthly Revenue" as total sales, including pending payments, whereas the **Operations Team** interprets it as completed and shipped orders only.

3. Interest Conflict

- The Customer Service Lead wants a "guest checkout" to reduce friction and increase immediate sales. The Data Analyst wants to force all users to create a detailed profile to collect long-term marketing data.

4. Value Conflict

- The **Founder** insists that the site must only promote eco-friendly, sustainable products to align with the company's principles. **The Investment Board** demands the inclusion of high-margin plastic goods to maximize short-term profit

5. Relationship Conflict

- The **Senior Developer** and the **UX Designer** have had a poor personal relationship since a previous project. Because of this, the Developer automatically rejects any requirement the Designer proposes for the checkout UI, leading to negative interpersonal behavior.

6. Structural Conflict

- The **Cybersecurity Department** (which has high organizational authority) mandates a 3-step authentication process. The **Mobile App Team** is forced to comply even though they know it will slow down their development schedule, as they are dependent on Security's approval.

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REQUIREMENTS NEGOTIATION

01 Agreement
Negotiate a solution

02 Compromise
Find compromise between alternatives

03 Voting
Vote for solution

04 Definition of variants
Allow systems variants

CONFLICT RESOLUTION TECHNIQUES



Overruling 05
Higher hierarchical power decides

Consider-all-facts 06
Investigate all influencing factors

Plus-minus-interesting 07
Evaluate positive / negative factors

Decision matrix 08
Evaluate alternatives with criteria

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CONFLICT RESOLUTION TECHNIQUES EXAMPLE

1 Agreement (Negotiate a solution)

Conflict: Marketing wants flat-rate shipping, while Logistics wants dimension-based costs (Subject Matter Conflict).

Resolution: The teams meet and negotiate a tiered flat-rate system. They agree on three fixed prices (Small, Medium, Large) based on standard box sizes, satisfying the need for simplicity while accounting for physical shipping costs.

2 Compromise

Conflict: Finance wants revenue to include pending payments, while Operations only wants completed orders (Data Conflict).

Resolution: They find a compromise by creating a dashboard that displays both "Gross Revenue" (all orders) and "Net Realized Revenue" (shipped orders). Neither side wins entirely, but both data needs are met.

3 Voting

Conflict: A team of 10 developers is split on whether to use a specific third-party payment gateway or build a custom one (Interest Conflict).

Resolution: After a formal presentation of both options, the project lead holds a vote. The solution with the majority (e.g., 7 out of 10) is selected as the final requirement.

4 Definition of Variants

Conflict: The UK branch wants a GDPR-compliant "Opt-In" for marketing emails, while the US branch wants an "Opt-Out" system to maximize sign-ups.

Resolution: Instead of choosing one, the team defines variants. The system is built with a localized configuration where the checkout page automatically switches its behavior based on the customer's IP address or region.

5 Overruling

Conflict: Mobile App developers want a fast 1-step login, but the Cybersecurity Lead demands 3-step authentication (Structural Conflict).

Resolution: Because the Cybersecurity Lead holds higher hierarchical power regarding system safety, they overrule the developers' preference to ensure the company remains compliant with security standards.

6 Consider-all-facts

Conflict: Choosing between two different cloud hosting providers for the store.

Resolution: The team investigates all influencing factors, such as uptime history, cost per gigabyte, customer support response times, and regional server availability, to determine which best meets the requirements best

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CONFLICT RESOLUTION TECHNIQUES EXAMPLE (cont.)

7 Plus-minus-interesting (PMI)

Conflict: The CEO wants to move to an "Eco-only" product line (Value Conflict).

Resolution: The board evaluates the positive, negative, and interesting factors:

Plus: Better brand reputation.

Minus: Loss of 40% of current high-margin inventory.

Interesting: Potential to attract a completely new demographic of premium shoppers.

8 Decision Matrix

Conflict: The Data Analyst wants forced registration, but Customer Service wants guest checkout (Interest Conflict).

Resolution: The team uses a decision matrix to evaluate alternatives with criteria like "Conversion Rate Impact," "Data Value," and "Implementation Time." By scoring "Guest Checkout" and "Forced Account" against these criteria, the option with the highest weighted score is selected objectively.

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Conflict Documentation

REQUIREMENTS NEGOTIATION

What should be documented

1. Assumptions concerning the conflict and its resolution
2. Potential alternatives considered
3. Constraints influencing the chosen technique and/or resolution
4. The way the conflict was resolved, including reasons for the chosen resolutions
5. Decision-makers and other contributors
6. Detected but not yet resolved conflicts in a TO-DO-List

Why document

1. To ensure traceability of positions and arguments.
2. To avoid that achieved resolutions are forgotten.
3. To facilitate their consideration in subsequent requirements engineering (development) activities.

Risks if conflict is not documented

- Handling conflicts repeatedly
- Inappropriate conflict resolution

Conflict Documentation - Example

REQUIREMENTS NEGOTIATION

Requirement Conflict Memo: CRM-2025-004
Conflict Identification:
Interest Conflict between Customer Service and the Data Analytics team regarding the checkout flow.

Assumptions Concerning Conflict:
1. Customer Service assumes that any barrier to checkout (like forced registration) will lead to abandoned carts and lost revenue.
2. Data Analytics assumes that without a persistent user ID, the company cannot accurately calculate Customer Lifetime Value (CLV) or run personalized remarketing campaigns.

Potential Alternatives Considered:
Alternative 1: Forced registration for all (Initial Data Analytics requirement).
Alternative 2: Guest checkout only (Initial Customer Service requirement).
Alternative 3: "Social Login" (Google/Facebook) to speed up registration.

Constraints:
1. The system must be launched by Q3, and the budget for third-party social login integration is limited.

Chosen Technique: Decision Matrix.

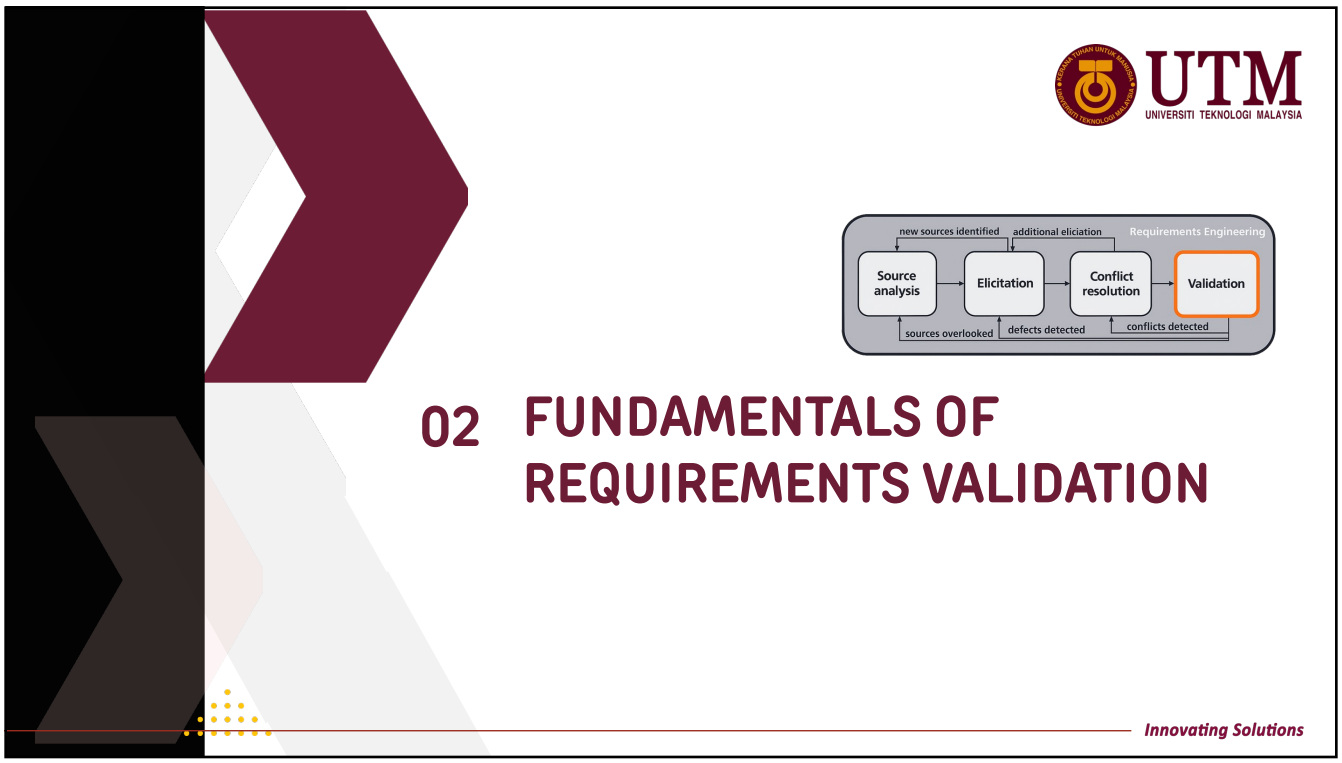
Reasoning:
This technique allowed the team to evaluate alternatives objectively based on weighted criteria: "Impact on Conversion Rate," "Data Richness," and "Implementation Cost."

Resolution Process & Outcome:
The team performed a Consider-all-facts investigation into industry benchmarks.

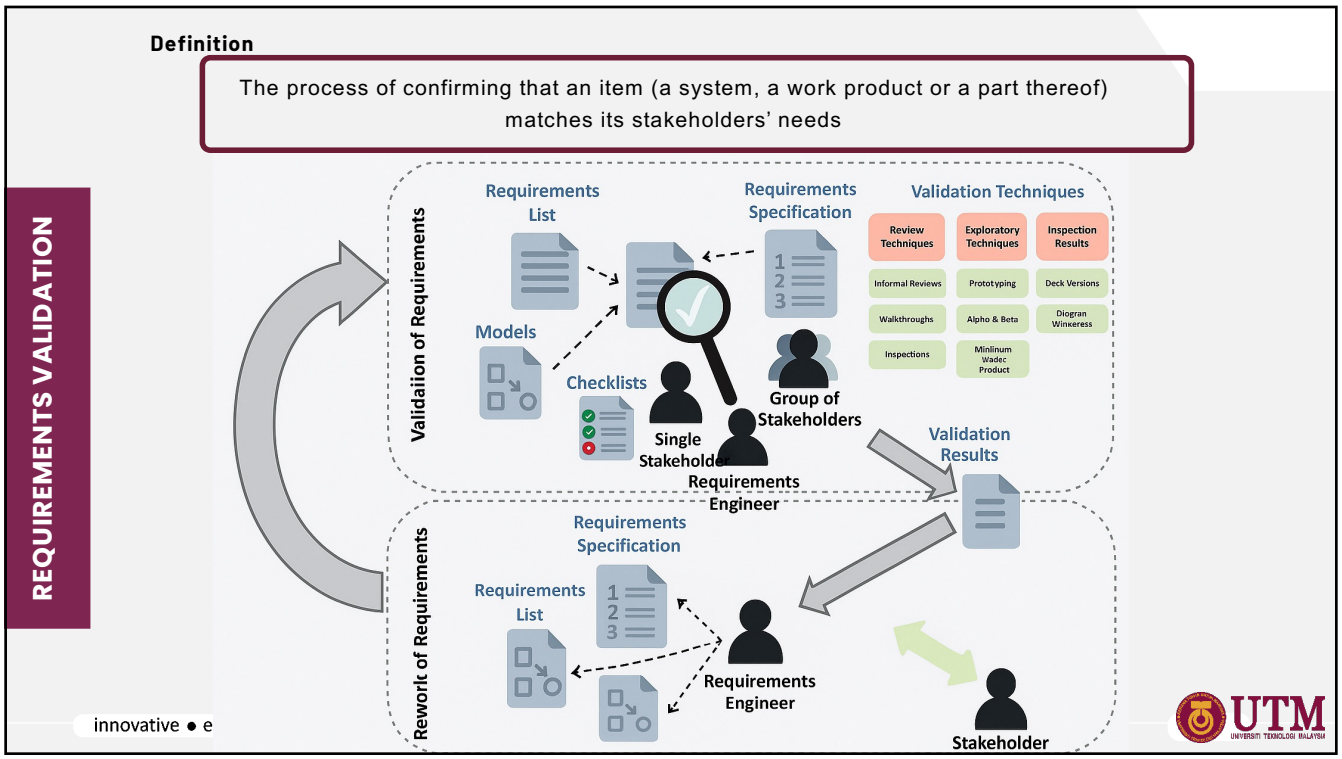
Decision:
The "Guest Checkout with Post-Purchase Registration" variant was selected. Users can checkout as guests, but are offered a discount coupon on the "Thank You" page if they create a password then.

Decision-Makers:
1. Chief Technology Officer (Final Sign-off)
2. Lead UX Designer (Contributor)
3. Marketing Director (Contributor)

TO-DO-List (Unresolved):
The legal team still needs to verify if the "Post-Purchase" data collection complies with updated regional privacy laws.


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new sources identified
additional elicitation
Requirements Engineering
Source analysis
Elicitation
Conflict resolution
Validation
sources overlooked
defects detected
conflicts detected
02 FUNDAMENTALS OF REQUIREMENTS VALIDATION
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Definition
The process of confirming that an item (a system, a work product or a part thereof) matches its stakeholders' needs
REQUIREMENTS VALIDATION
Validaiion of Requirements
Requirements List
Requirements Specification
Validation Techniques
Review Techniques
Exploratory Techniques
Inspection Results
Informal Reviews
Prototyping
Deck Versions
Walkthroughs
Alpho & Beta
Diogran Winkeress
Inspections
Minlinum Wadec Product
Models
Checklists
Group of Stakeholders
Single Stakeholder
Requirements Engineer
Validation Results
Rework of Requirements
Requirements Specification
Requirements List
Requirements Engineer
Stakeholder
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Validate Quality Criteria

REQUIREMENTS VALIDATION

Quality Criteria for Single Requirements

- Unambiguous
- Necessary
- Adequate
- Complete
- Verifiable
- Understandable

- most important quality criteria.
- Without them, a requirement is useless regardless of the fulfillment of all other criteria

Quality Criteria for Work Products

- Consistent
- Non-redundant
- Complete
- Modifiable
- Traceable
- Conformant

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Important Aspects of Validation

REQUIREMENTS VALIDATION



1 Involvement of the right stakeholders

- Independence of the auditor
- Involve external ones, when internal lack of expertise

2 Separating the identification and the correction of defects

- Focusing on error identification increases detection rate.
- Avoidance of error correction decreases unproductive activity

3 Validation from different views

Client view, developer view, auditor view, RE view

4 Adequate change of documentation type

- Play out the strengths, compensate weaknesses
- Transcribing from one into another form helps identifying errors

5 Construction of development artifacts

Simultaneous activities of developers and testers

6 Repeated validation

Re-validate constantly in a heavily changing environment

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Validation Techniques

REQUIREMENTS VALIDATION



Review Techniques

- Informal Reviews
- Walkthrough
- Inspection



Exploratory Techniques

- Prototyping
- Alpha and Beta Testing
- A/B Testing
- Minimum Viable Product



Sample Development

- Sample Development

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Validation Techniques

REQUIREMENTS VALIDATION

Techniques differ in formality and effort

Factors influencing selection of technique

01 Software development life cycle model

02 Maturity of the development process

03 Complexity and risk level of the system

04 Legal or regulatory requirements

05 Need for an audit trail

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Validation Techniques: Review

REQUIREMENTS VALIDATION

- Rely on visual study of early and intermediate work products.
- Range from informal to very fomal
- Can be applied from the very beginning of a project until the implementation of the system
- In most cases, reviewing the requirements is limited to the earlier phases of a project.
- Typically, static work products checked that define or describe how the system should work.
- Principle of reviews:
  - Four eyes see more than two
    - especially if they were not involved in producing the checked item.
  - Early error detection
    - while they can be removed at a relatively low cost

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Review Techniques

VALIDATION TECHNIQUES: REVIEW



Inspections
- Formal and strict review process with defined roles
- Typically uses checklists

Walkthroughs
- Lightweight review with less rules
- Reviewers study requirements in advance
- Discussion of the identified errors in quality during a group session

Informal Reviews
- Other person (e.g. co-worker) reviews work products
- Small group of team members, peers, and/or users

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Review Techniques: Informal Reviews

VALIDATION TECHNIQUES: REVIEW

- Typically small group of team members, peers, and/or users involved in the project
- Composition of group may be prescribed by company regulation

Send a work product to a group of people with the request to validate it

Collect all review comments after a short (but often not predefined) period

Update the work product at hand

Document the comments in a review register

Keep track of the way in whhich comments are processed

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Review Techniques: Informal Reviews

VALIDATION TECHNIQUES: REVIEW

- Typicall you are free to decide whether and how to use the comments (informal nature)
- Repeat review over several drafts versions until satisfied with the quality.
- Works well, if participants are:
  - committed to quality
  - able and willing to spend enough time on the review process.
- Easy, cheap, and approachable means of validation.
- Common for early drafts.
- For the final version of a work product, a more formal technique may be a better choice.

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Review Techniques: Walkthroughs

VALIDATION TECHNIQUES: REVIEW

- 2 variants of walkthrough:
  a. reviewers join the meeting without any preparation and listen to the author, asking adhoc questions.
  b. reviewers obtain the work product before the meeting and will prepare questions for the author



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Review Techniques: Walkthroughs

VALIDATION TECHNIQUES: REVIEW

- Best to apply:
  - in an early project phase to discuss the feasibility of a certain system concept or solution outline.
  - on the transfer of an intermediate work product to another party who will use it as input for a subsequent development.
- In iterative projects, walthroughs are mostly present in the form of:
  - regular refinement sessions prior to an iteration
  - sprint reviews at the end of it.
- Useful and highly visual via storyboarding, workflows etc.
- Seen as a more passive approach presenting the information finding defects but not proposing solution.
- Opportunity for reviewers analyse and query the document.

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Review Techniques: Inspection

VALIDATION TECHNIQUES: REVIEW

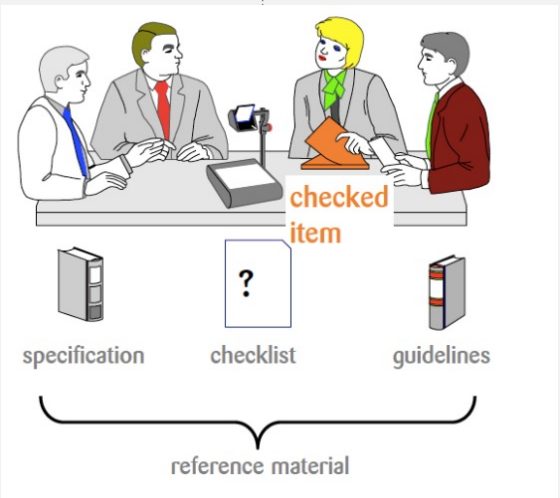
- Most formal review technique
- Responsibility for the review lies not with the author but with an independent review leader, often called moderator.
- Normally performed in the form of a meeting with the moderator, the author and a group of inspectors.
- Inspectors are selected from peers, business, users and/or experts
- Inspectors are asked to
  - check the work product based on their specific expertise
  - verify its adherence to applicable standards, norms and regulations
  - evaluate it against agreed objectives.

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Review Techniques: Inspection

VALIDATION TECHNIQUES: REVIEW

- Inspectors perform check during thorough individiual preparation prior to the actual meeting
- Inspectors are often guided by detailed checklists
- Author participates in the review meeting in the role of a listener
  - explaining things that are not clear
  - trying to understand the comments of the inspectors and the consequences for the work product.
- Inspections follow a strict and documented process that is managed by the moderator



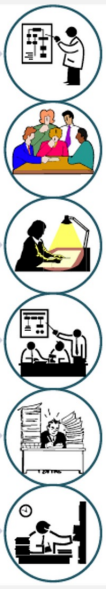
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Review Techniques: Inspection

VALIDATION TECHNIQUES: REVIEW

With a formal review, there are 6 basic steps to execute:

1. Planning
2. Kick-Off
3. Individual Preparation
4. Examination
5. Rework
6. Follow Up



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Review Techniques: Participants in Inspection

VALIDATION TECHNIQUES: REVIEW

1. Manager

- Decides what to review
- Allocate time
- Determine whether objectives have been met.

2. Moderator

- Lead and manage the team, including planning, motivating, facilitating meetings, training reviewers and monitoring the process.
- Records summary results for follow-up action

3. Author

- Prepares the material to be reviewed,
- Participates in discussions during the review meeting
- Incorporates the agreed changes afterwards

4. Reviewer

- Inspect documents and Responsible for finding defects.
  - Stakeholders should have a technical or business background. Ideally from different areas and focuses, such as Business users, developers, analysts, IT operations, testers and compliance officers.

5. Scriber

- Record and documents all the issues, problems and open points that were identified during the meeting.

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Review Techniques: Composition of the team in Inspection

VALIDATION TECHNIQUES: REVIEW

Inspectors are representatives of all stakeholder groups

- Representative of the area of expertise
  - Represent the knowledge in the application domain of the planned system
  - Necessary to check
    - The content, for validation
    - The agreement
- Domain Expert
  - Know the specifics of the application area
  - Support the checking of the content, the validation
- Representative of the development team
  - Architects, designer, developer, tester
  - check
    - The documentation
    - For feasibility, testability and the quality requirements
- Representative of the Requirements Engineers
  - Not involved, checks the "handicraft"



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Review Techniques: Inspection

VALIDATION TECHNIQUES: REVIEW

- Focus is on:
  - finding defects and measuring defined quality aspects
  - providing a detailed audit trail
- Inspections are often used to decide on the release of a work product for a next step in the development process, or for final implementation.
- Inspections are mostly applied in (safety) critical systems and business process.
- In agile approaches, this formal way of reviewing is incorporated in the methodology itself
  - with the Scrum ceremonies (refinement, planning, sprint review)

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Review Techniques: Inspection

VALIDATION TECHNIQUES: REVIEW

Reviews
The resulting "review report"

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Classification of findings

- good
- critical error
  - must be fixed
  - no approval of the checked item without further review
- major defect
  - should be fixed
  - no approval without update of the checked item
- minor defect
  - grammar, layout, …
  - the checked item can be approved with / without update
- open issue
  - no clarification or agreement reached in the review meting
  - no approval of the checked item without clarification

Recommendation for decision about the checked item

- accept
  - with or without update
- reject
  - update is compulsory
  - re-review is mandatory

Zühlke Engineering AG — **Review Template**

Review Object
ID / Title / Version: _____
Author: _____
Review Targets: _____

Review Result
[ ] accepted as is
[ ] accepted with changes
[ ] rejected (new review needed)

Meeting Information
Date / Time: _____
Participants: _____
(with roles in review)

overall preparation _____
time of reviewers [h]

Review Findings — [1] Type: good, critical, major, minor, open issue

Nr.	ID/Page	Reviewer Comment	Type[1]	Reviewer	Additional comments	Done
1						
2						
3						
4						

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Review Techniques: Inspection

VALIDATION TECHNIQUES: REVIEW

Example Review Report of
Use Case Diagram and Specification

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Zühlke Engineering AG — **Review Template**

Review Object
ID / Title / Version: ECP Use Case Document
Author: John Smith
Review Targets: Review content and document aspect

Review Result
[ ] accepted as is
[ ] accepted with changes
[X] rejected (new review needed)

Meeting Information
Date / Time: 24.12.2013, 9:00-17:00h
Participants: M. Lampe (Moderator), R. Lock, J. Smith, A. Brown (Inspectors)
(with roles in review)

overall preparation 2h
time of reviewers [h]

Review Findings — [1] Type: good (G), critical (C), major (MJ), minor (MN), open issue (OI)

Nr.	ID/Page	Reviewer Comment	Type[1]	Reviewer Name	Additional comments	Done
1	UC Diagram	Use Case "Manage ECP" and actor "System Admin" are missing	C	R. Lock		
2	UC Diagram	Use Case name UC10 "Ordering" should be renamed to "Order Products" or "Checkout basket"	MJ	J. Smith		
3	UC Diagram	Include relationship between UC10 and UC13 not correct UML notation	MN	J. Smith		
4	UC10	not all actors are listed, credit card clearing system missing	MJ	A. Brown, J. Smith		
5	UC10	Precondition is formulated as an action and not as a condition	MN	R. Lock		
6	UC10	If-else conditions do not make sense, better use alternative scenarios	MJ	A. Brown		
7	UC10	Scenario step 4 takes design decision already ahead, reformulate in a more genearl way	MN	J. Smith		
8	UC13	Unclear which data the credit card clearing system needs	OI	R. Lock	expert is E. Meyer	
9	UC13	Spelling error in step 4: "recept"->"receipt"	MN	J. Smith		
10	UC13	Post-condition is missing, should check if payment is really processed	C	R. Lock, A. Brown	expert is E. Meyer	

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Review Techniques: Inspection

VALIDATION TECHNIQUES: REVIEW

Example: Checklist for Review of Use Cases (1/2)

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All questions need to be answered with "yes"

Topic	Question
Use Case Title	1. Is the title an active phrase expressing the goal of the primary actor? 2. Can the goal be achieved by the system?
Short description	3. Does the description clearly describe the goal of the primary actor? 4. Does it deliver an overview of the main activities?
Primary actor	5. Is the use case triggered by at least one (primary) actor? 6. Is the primary actor uniquely identified?
Prerequisites	7. Can the system fulfill the required prerequisites? 8. Is it true that no checking of the prerequisites is done in the use case?
Post condition	9. Can the post conditions always be met after a successful development?
Primary path	10. Is it described with 3 to 9 steps? 11. Do the steps lead from trigger to achieved post condition? 12. Are branches and loops correctly described? 13. Are exceptions described as such? 14. At conditions in the flow: Is it defined what procedure shall be taken if the condition is not fulfilled?

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Review Techniques: Review Report Example

VALIDATION TECHNIQUES: REVIEW

Table 3.2 Problem log template

No.	Page/line	Defect category	Severity	Defect type	Source of defect	Description, comments
...	.....	.....	.....	.....	.....	.....

Defect categories: Missing, Defect, Redundancy

Table 3.3 A problem log filled out by one of the reviewers

No.	Page/line	Defect category	Severity	Defect type	Description, comments
1	Fig. 1	Defect	Small	Syntax	"Electric lighnting" instead of "Electric lightning"
2	Fig. 1	Defect	Large	Logic	The element following "Electric Lightning" should be the "Electric Storm" skill, not "Firestorm"
3	Fig. 1	Missing	Large	Logic	The skill "Blinding the opponent" is missing, the introduction of which to the skill tree was agreed at the Feb 4th meeting

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Validation Techniques: Exploratory Techniques

REQUIREMENTS VALIDATION

- Exploratory techniques are dynamic
- Look at the (actual or simulated) behaviour of the operative system as experienced by the users through the user interfaces
- Participants are invited to use the prototype / system in a way that is similar to the intended use in production
- Findings include defects and suggestions for improvement
- Common in iterative and design thinking development approaches.

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Exploratory Techniques: Prototyping

REQUIREMENTS VALIDATION: EXPLORATORY TECHNIQUES

In validation a specific early version of the system is given to group of stakeholders for evaluation

- May be explicitly built
  - for validation purposes, after which it is n discarded (exploratory or throwaway prototype)
  - to check a specific characteristic, such as user interface, security, or performance
- Evolutionary prototypes can also be used for validation
  - continuously updated and extended until they end up in the final product (see MVP)

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Exploratory Techniques: Prototyping

REQUIREMENTS VALIDATION: EXPLORATORY TECHNIQUES

- Group of stakeholders and prospective users gain hands-on experience with (part of) the system under development
- From the outside prototype looks like the intended system
- Internal structure may still be unfinished, inoperative, or even completely missing
- With prototyping, validation and elicitation go together

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Exploratory Techniques: Alpha and Beta Testing

REQUIREMENTS VALIDATION: EXPLORATORY TECHNIQUES

- End users test a fully featured, completely working pre-production versin of the system in operation.
- End users validate its look and behaviour
- Useful for checking certain assumptions made during elicitation and development
- User feedback gives valuable clues to the quality of the requirements.

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Exploratory Techniques: Alpha and Beta Testing

REQUIREMENTS VALIDATION: EXPLORATORY TECHNIQUES

Alpha testing

- At the developer's site in a simulated environment
- Small group of participants
- Some guidance may be given
- Interaction of the users with the system is observed (e.g. in a usability lab)

Beta testing

- At the end user's sites in real production (or in environment the end users decide)
- Stimulate all participants to give their feedback and to provide an easy way to do so
- System is offered (mostly) for free
- Sometimes selected byt usually unknown group of users.

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Exploratory Techniques: A/B Testing

REQUIREMENTS VALIDATION: EXPLORATORY TECHNIQUES

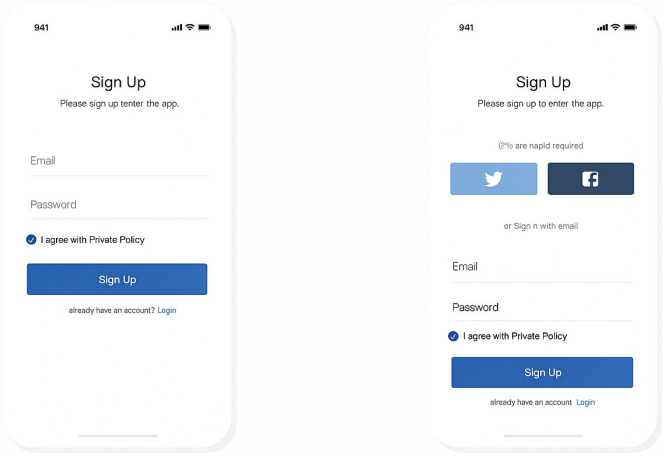
- Performed with a released version of the system in the fully operational enviroment.
- Can also be applied with pre-release versions in a protected test environment
- Split users randomly in two groups
- Offer system to users in two variants A and B
  - variants differ in design or functionality
  - variants realize the user goals in a different way
- Measured and compare reaction of both groups
- Groups need to be large enough to allow for statistical analysis
- Provides information on the quality of the underlying requirements and on the correctness of previous assumptions
- A/B tesing has a prominen role in the Lean Startup, one the design thinking approaches.

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Exploratory Techniques: A/B Testing

REQUIREMENTS VALIDATION: EXPLORATORY TECHNIQUES

Variant A: Signup with email **Variant V**: Signup with social media or email



Measure: Number of signups per day/week per variant

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Exploratory Techniques: Minimum Viable Product (MVP)

REQUIREMENTS VALIDATION: EXPLORATORY TECHNIQUES

- Version of a product with just enough features to be usable by early customers who can then provide feedback for future product development.
- Used in incremental development especially in agile development
- Used product cycle of build, measure, learn
  - Build and release a MVP
  - Measuring market reactions
  - Adjusting the system accordingly
- Approach can be seen as an exploratory validation of the requirements in production

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Exploratory Techniques: Minimum Viable Product (MVP)

REQUIREMENTS VALIDATION: EXPLORATORY TECHNIQUES

Sprint, Release and MVP

The core functionalities are developed first, allowing users to begin using and testing the product in real-world conditions.



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Exploratory Techniques: Minimum Viable Product (MVP)

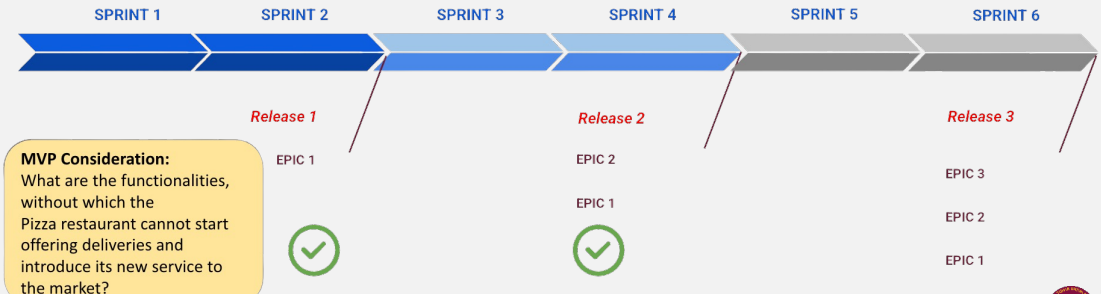
REQUIREMENTS VALIDATION: EXPLORATORY TECHNIQUES

MVP Example

Epic 1: Build Basic Website Interface

Epic 2: Implement Online Payment functionality

Epic 3: Enable Custom Pizza Creation — Not Critical for the website to start working



MVP Consideration:
What are the functionalities, without which the Pizza restaurant cannot start offering deliveries and introduce its new service to the market?

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Requirements Validation Techniques: Sample Development

REQUIREMENTS VALIDATION



Process:
- Requirements are provided as input to developers.
- Developers produce intermediate work products (for example design, codes, test cases, manuals)



Nature of validation:
- Validation is static, similar to review



Flaws of Detection
- Unclarities
- Omissions
- Inconsistencies



Quality Indication
- Number & severity of detected flaws = indicator of requirements quality

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THANK YOU

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